

6.2.2.2. Encoding of Vendor ID and Product ID in subject and issuer fields

The following subsections contain references to [VendorID](#) and [ProductID](#):

- [Section 6.2.2.3, “Device Attestation Certificate \(DAC\)”](#)
- [Section 6.2.2.4, “Product Attestation Intermediate \(PAI\) Certificate”](#)
- [Section 6.2.2.5, “Product Attestation Authority \(PAA\) Certificate”](#)

The values for VendorID and ProductID, where possible or required in [issuer](#) or [subject](#) fields SHALL be encoded by only one of two methods, without mixing the methods within a given field:

1. The "preferred method", using Matter-specific [RelativeDistinguishedName](#) attributes:
 - a. VendorID encoded as [AttributeTypeAndValue](#) entry with [type](#) equal to [1.3.6.1.4.1.37244.2.1](#), and value respecting encoding specified in [Section 6.1.1, “Encoding of Matter-specific RDNs”](#).
 - b. ProductID encoded as [AttributeTypeAndValue](#) entry with [type](#) equal to [1.3.6.1.4.1.37244.2.2](#), and value respecting encoding specified in [Section 6.1.1, “Encoding of Matter-specific RDNs”](#).
2. A "fallback method" to support certificate authorities that only allow customary [RFC 5280](#) OIDs in the arc [{joint-iso-itu-t\(2\) ds\(5\) attributeType\(4\)}](#) for [type](#) values in [AttributeTypeAndValue](#) entries of [RelativeDistinguishedName](#) elements is to encode them as substrings within the [commonName](#) attribute type ([{joint-iso-itu-t\(2\) ds\(5\) attributeType\(4\) commonName\(3\)}](#)):
 - a. VendorID value encoded with substring [Mvid](#): followed by exactly 4 uppercase hexadecimal characters without elision of leading zeroes, anywhere within the [commonName](#), such as for example:
 - i. VendorID 0xFFFF1 (65521 decimal): [Mvid:FFF1](#)
 - ii. VendorID 0x2A (42 decimal): [Mvid:002A](#)
 - b. ProductID value encoded with substring [Mpid](#): followed by exactly 4 uppercase hexadecimal characters without elision of leading zeroes, anywhere within the [commonName](#), such as for example:
 - i. ProductID 0xC20A (49674 decimal): [Mpid:C20A](#)
 - ii. ProductID 0x3A5 (933 decimal): [Mpid:03A5](#)
 - c. For both VendorID and ProductID, the following additional rules apply to the "fallback method":
 - i. The leftmost match having a correct encoding SHALL be used, with other correct matches discarded.
 - ii. For either of VendorID and ProductID, if the prefix string is found on its own anywhere within the [commonName](#), but there is no fully correct match anywhere in the [commonName](#), the field SHALL be considered incorrectly formatted.

The "preferred method" leaves more space for content in the [commonName](#) attribute type if present. It is also less ambiguous which may allow simpler processing of certificate issuance policy validations in CAs that support the Matter-specific [RelativeDistinguishedName](#) attributes, and simplify the audit of certificates where Vendor ID and Product ID appear.

The "fallback method" is present to support less flexible CA infrastructure.